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Preamble

Jack code written in 2025 and translated into english in 2026.

Isolation tests

Memory

The proposed `Memory.jack` source file implements a first-fit strategy for the improved memory allocation algorithm (see § 12.1.3). Here is how to test it :

1. Recompile Jack source files :

```
c:\(...)\nand2tetris\projects\12>call ..\..\tools\JackCompiler.bat Perso\MemoryTest  
compiling "c:\(...)\nand2tetris\projects\12\Perso\MemoryTest"
```

2. Open the VM Emulator : `call ..\..\tools\VMEmulator.bat`

File / Load Script / Select `c:\(...)\nand2tetris\projects\12\Perso\MemoryTest\MemoryTest.tst`

Select Animate / No animation to get a fast answer.

3. Run the test (`>>`), choose *Yes* for the Confirmation Message, click *OK* when execution has finished.

4. Check that *End of script - Comparison ended successfully* appears at the bottom of the screen.

As a complementary test, the same procedure can be applied to `Perso\MemoryTest\MemoryDiag`.

However, I do not remember having done a step-by-step debugging in the VM Emulator to fully test the `Memory.alloc()` and `Memory.deAlloc()` methods.

Array

Same 4 steps as for the *Memory* isolation test.

Math

Again, same 4 steps as for the *Memory* isolation test with one more verification. If step 4 has succeeded, check the memory :

- Go to the 8000 memory location using the binocular tool.
- Check that `RAM[8000 → 8013]` contains the values described in `MathTest\Main.jack`.

String

Here, the test procedure changes a bit. We have the same 3 first steps as the *Memory* isolation test but the fourth is different. Here, we have to check that the screen output is identical to the screen capture in the documentation, that is :

```
new,appendChar: abcde
setInt: 12345
setInt: -32767
length: 5
charAt[2]: 99
setCharAt(2,'-'): ab-de
eraseLastChar: ab-d
intValue: 456
intValue: -32123
backSpace: 129
doubleQuote: 34
newLine: 128
```

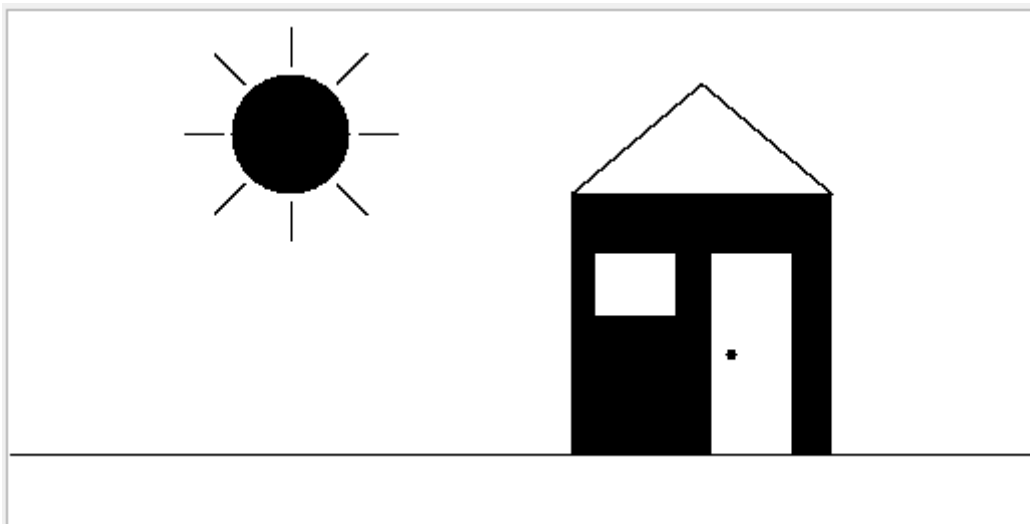
Output

Same procedure as for the previous test. The result has to be identical to this :

```
A
0123456789
ABCDEFGHIJKLMNOPQRSTUVWXYZ abcdefghijklmnopqrstuvwxyz
!#$%&'()*+,-./:;<=>?@[ ]^_`{|}~"
-12346789
C                                     D
```

Screen

Same test again. The result has to be identical to this :



Keyboard

Same test but some typing to do this time. Expected result :

```
keyPressed test:
Please press the 'Page Down' key
ok
readChar test:
(Verify that the pressed character is echoed to the screen)
Please press the number '3': 3
ok
readLine test:
(Verify echo and usage of 'backspace')
Please type 'JACK' and press enter: JACK
ok
readInt test:
(Verify echo and usage of 'backspace')
Please type '-32123' and press enter: -32123
ok
Test completed successfully
```

Sys

Similar test except we have to check that ~ 2 seconds have passed between the moment we press a key and the moment the *Time is up* message appears on the screen :

```
Wait test:
Press any key. After 2 seconds, another message will be printed:
Time is up. Make sure that 2 seconds elapsed.
```

Global test

The global test should be run once the 8 isolation tests are ok. This final test is done with the *Pong* game :

